

ENEL700 Tutorial 1 - Solutions

1. List five types of media used for communication, and state which three are the most commonly used.

Electrical conductors (twisted pair, coaxial cable); Optical media (fibre); Free-space (wireless); and water.

The underlined media are the three most widely used.

2. What piece of equipment acquires a signal from a communication medium and recovers the original information signal?

Demodulator

3. What is a baseband signal? And a broadband (passband) signal?

Baseband signal: original information signal that can be sent unmodified or used to modulate a carrier for transmission over a medium

Broadband signal: high-frequency carrier signal that has been modulated by audio, video, or data.

4. What is the term used to describe two-way communication in which each party takes turns transmitting? Give three examples?

Half-duplex

Police/Military radio transmission; CB radio; Amateur radio.

5. In what segment of the spectrum do TV channels 2 to 13, and FM broadcasting, appear?

VHF (very high frequencies)